

RegVelo: Gene-regulatory-informed Dynamics of Single Cells

01. Paper Information

Paper	RegVelo: gene-regulatory-informed dynamics of single cells
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02. At a Glance

No.	Aspect	Reading note
01	Question	How can we infer single-cell dynamics while explicitly modeling the gene regulatory interactions that drive transcriptional state changes and fate decisions?
02	Input	Spliced and unspliced single-cell RNA abundance, together with prior gene regulatory network information from sources such as ATAC-seq, multiome measurements, motif databases, ChIP-seq resources, or curated regulatory knowledge.
03	Model	RegVelo is a Bayesian deep generative model that extends RNA velocity by coupling genome-wide splicing kinetics through a GRN-informed transcription predictor. Transcription is modeled as a time- and regulation-dependent process, yielding posterior velocity distributions, latent time, kinetic parameters, and an inferred GRN.
04	Output	Cell- and gene-specific latent time, posterior RNA velocity, uncertainty estimates, refined regulatory edges, fate probabilities through CellRank integration, lineage-driver rankings, and in silico regulon perturbation predictions.
05	Main claim	By integrating splicing dynamics with gene regulatory structure, RegVelo produces more faithful dynamic inference, more mechanistic driver discovery, and actionable perturbation simulations across development and differentiation systems.
06	Why it matters	The key conceptual move is to turn RNA velocity from a gene-independent kinetic model into a regulatory dynamical system. This links local velocity fields, long-term fate probabilities, and causal hypotheses about transcription factor perturbations.

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1 One-sentence summary

Core summary

RegVelo couples RNA splicing kinetics with gene regulatory networks in a Bayesian generative model, enabling uncertainty-aware velocity inference, regulatory circuit discovery, CellRank-based fate prediction, and in silico perturbation analysis of transcription factor regulons.

2 Why this paper matters

RNA velocity methods infer a local direction of cell-state change from unspliced and spliced RNA. This is powerful, but most velocity models treat genes as independent dynamical systems with constant or weakly time-varying transcription rates. This creates a conceptual gap: cell fate transitions are controlled by regulatory circuits, yet the velocity model often ignores the gene-gene interactions that produce those dynamics.

On the other side, gene regulatory network inference methods can identify putative transcription factor-target relationships, but they often do not model cellular dynamics directly. As a result, regulation and dynamics are frequently connected only after the main analysis. RegVelo tries to bridge this gap by making transcription itself a regulated process inside the velocity model.

Key takeaway

RegVelo is important because it makes RNA velocity more mechanistic. Instead of asking only “where is this cell moving?”, the model asks “which regulatory system generates this motion, and what happens to fate if we perturb that regulatory system?”

3 Biological and computational background

3.1 The biological problem

Cell fate decisions are controlled by regulatory programs. Transcription factors activate or repress target genes, and these interactions shape trajectories such as endocrine differentiation, hematopoiesis, neural crest development, and pigment cell specification. A model that only estimates velocity vectors can identify directions of change, but it may not explain why cells move in those directions.

The paper targets three connected problems:

- **Dynamic inference:** estimate the direction and temporal ordering of cell-state change.
- **Regulatory inference:** infer which transcription factors regulate which target genes.
- **Perturbation prediction:** simulate how perturbing a regulator changes local dynamics and long-term fate probabilities.

Why RegVelo is needed

Problem 3.1

Classical RNA velocity can fail when transcription rates are not constant or when gene dynamics are strongly coupled by regulation. GRN inference alone can suggest regulatory edges, but it does not automatically tell how perturbing those edges changes cell fate. RegVelo connects these two layers in one model.

3.2 The computational framing

RegVelo models a high-dimensional dynamical system in which transcription rates depend on regulatory inputs. For each gene g , the unspliced and spliced abundances evolve as:

$$\frac{du_g(t)}{dt} = \alpha_g(t) - \beta_g u_g(t), \quad (1)$$

$$\frac{ds_g(t)}{dt} = \beta_g u_g(t) - \gamma_g s_g(t), \quad (2)$$

where u_g is unspliced RNA, s_g is spliced RNA, α_g is transcription, β_g is splicing, and γ_g is degradation. The main difference from standard velocity models is that RegVelo makes transcription regulatory:

$$\alpha_g(t) = h([Ws(t) + b]_g), \quad (3)$$

where W is the learned regulatory weight matrix, b is a basal transcription term, and h is a nonlinear activation function.

Method note

The key modeling idea is that genes are no longer independent one-dimensional ODEs. They are coupled through a GRN-informed transcription function. This makes the velocity field depend on regulatory circuitry rather than only on each gene's own splicing kinetics.

4 Method breakdown

4.1 Step 1: encode spliced and unspliced counts

RegVelo takes the spliced and unspliced count matrix as input. A neural encoder maps each cell into a latent representation. A decoder then uses this representation to infer cell- and gene-specific latent time. This builds on the veloVI-style variational inference framework, but introduces an explicit regulatory component.

4.2 Step 2: build a GRN-informed transcription predictor

The model represents the GRN as a weight matrix W . Each entry W_{rg} describes how regulator r affects target gene g :

- positive weight: activation;
- negative weight: repression;
- zero or near-zero weight: no inferred regulatory effect.

Prior GRN information can come from multiome data, ATAC-derived motif links, ChIP-seq resources, or public regulatory databases. During training, RegVelo refines the prior network by learning edge weights and potentially correcting incomplete or noisy prior information.

4.3 Step 3: solve coupled genome-wide dynamics

Unlike classical RNA velocity models that solve gene dynamics independently, RegVelo solves a coupled ODE system. The model jointly learns transcription, splicing, degradation, latent time, and regulatory weights. Because the ODE system is nonlinear and genome-wide, the solution is computed numerically rather than through a closed-form single-gene expression.

Method note

This is the main technical departure from scVelo-style dynamical models: RegVelo treats transcription as a function of the current cell state and regulatory network, so the dynamics of one gene can influence the dynamics of another gene.

4.4 Step 4: infer velocity and uncertainty

After training, RegVelo outputs posterior velocity distributions. This means the model does not only give one velocity estimate; it can also quantify uncertainty in the inferred direction of cell-state change.

The paper distinguishes two intuitive types of uncertainty:

- **Intrinsic uncertainty:** uncertainty in the local direction of change along the phenotypic manifold.
- **Extrinsic uncertainty:** uncertainty in future cell states generated from the posterior dynamics.

These uncertainty estimates are important because velocity inference is noisy, especially in sparse single-cell data.

4.5 Step 5: connect velocity to fate probabilities with CellRank

RegVelo produces a velocity field. To analyze long-term cell fate, the paper integrates RegVelo with CellRank. CellRank converts local velocity and expression similarity into a Markov transition matrix and then computes terminal states and fate probabilities.

For a terminal fate k , the resulting fate probability is:

$$F_{ik} = P(\text{cell } i \text{ eventually reaches terminal state } k). \quad (4)$$

This connects local regulatory dynamics to global fate outcomes.

Key takeaway

RegVelo by itself gives a regulatory velocity field. RegVelo plus CellRank gives a fate map. This is why the method can simulate how perturbing a transcription factor affects not only immediate cell motion but also long-term terminal fate probabilities.

4.6 Step 6: simulate regulon perturbations

For in silico perturbation, RegVelo masks or removes a regulator's downstream targets from the GRN, recomputes the velocity field, and compares the perturbed dynamics with the original dynamics. Local changes can be measured by velocity dissimilarity, while global changes can be measured by CellRank fate probability shifts.

A simplified perturbation workflow is:

$$W \rightarrow W^{(-r)} \rightarrow v^{(-r)}(x) \rightarrow F^{(-r)}, \quad (5)$$

where $W^{(-r)}$ is the GRN after depleting regulator r 's regulon, $v^{(-r)}$ is the perturbed velocity field, and $F^{(-r)}$ is the perturbed fate probability matrix.

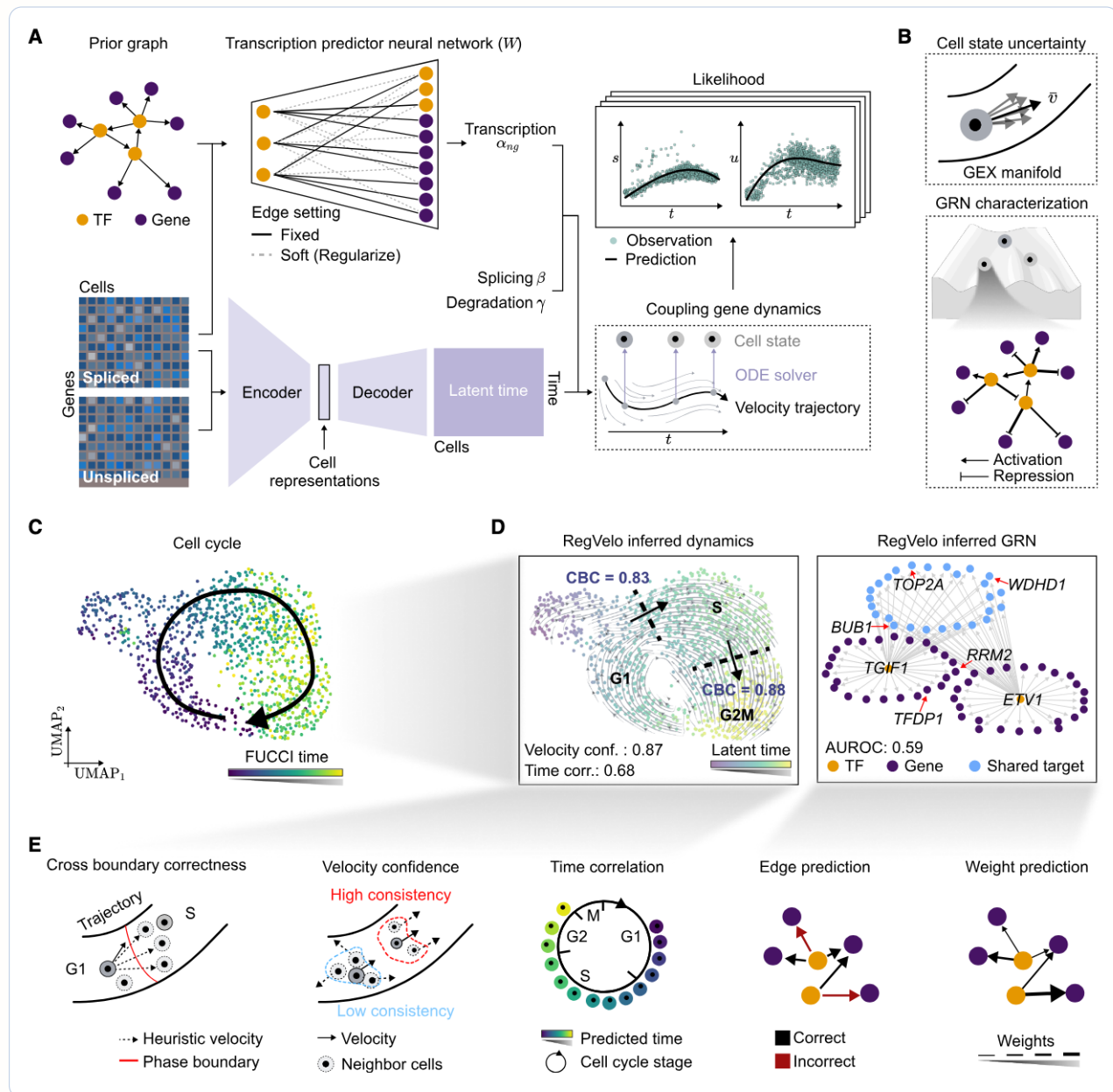


Figure 1: Main text Fig. 1A-E. RegVelo jointly models single-cell dynamics and gene regulatory networks. Use this figure to show how spliced/unspliced RNA, a prior regulatory graph, neural networks, and coupled ODE dynamics produce latent time, velocity, uncertainty, and an inferred GRN.

How to read Fig. 1

Figure note 4.1

Panel A is the central method diagram. It shows the two information streams: RNA dynamics are encoded into latent time, while prior regulatory knowledge constrains the transcription predictor. Panels C-E provide an intuitive benchmark on the cell cycle, where the ground-truth direction is known from Fucci markers.

5 Main experiments and evidence

5.1 Simulation and cell-cycle benchmarking

The paper first evaluates whether RegVelo can recover dynamics and regulation in settings where the direction of change is either simulated or approximately known. In simulated GRN systems, RegVelo is compared

with velocity and GRN inference baselines using velocity accuracy, latent time recovery, and regulatory edge prediction.

The real-world benchmark uses FUCCI cell-cycle data. FUCCI provides a protein-derived cell-cycle score, which serves as a proxy for ground-truth progression. RegVelo recovers the expected unidirectional G1-S-G2M flow, produces a coherent velocity field, and infers a GRN containing cell-cycle-related regulators and targets.

Key takeaway

The cell-cycle benchmark is useful because it is one of the rare single-cell settings where the expected transition direction is well constrained. RegVelo's advantage is not only better velocity, but also an inferred regulatory network that aligns with cell-cycle biology.

5.2 Pancreatic endocrinogenesis and in silico regulon perturbation

RegVelo is then applied to pancreatic endocrine development. A prior GRN is constructed using cis-regulatory elements, gene expression, and chromatin accessibility from a related multiome dataset. Passing RegVelo velocities to CellRank recovers the major terminal endocrine fates: alpha, beta, delta, and epsilon cells.

The perturbation analysis asks whether removing a transcription factor regulon changes fate probabilities in biologically meaningful ways. For example, the model recovers known lineage drivers such as *Arx*, *Pdx1*, and *Hhex*. It also prioritizes *Neurod2* for epsilon cell differentiation and suggests that the *Neurod2-Rfx6* relationship contributes to epsilon maturation.

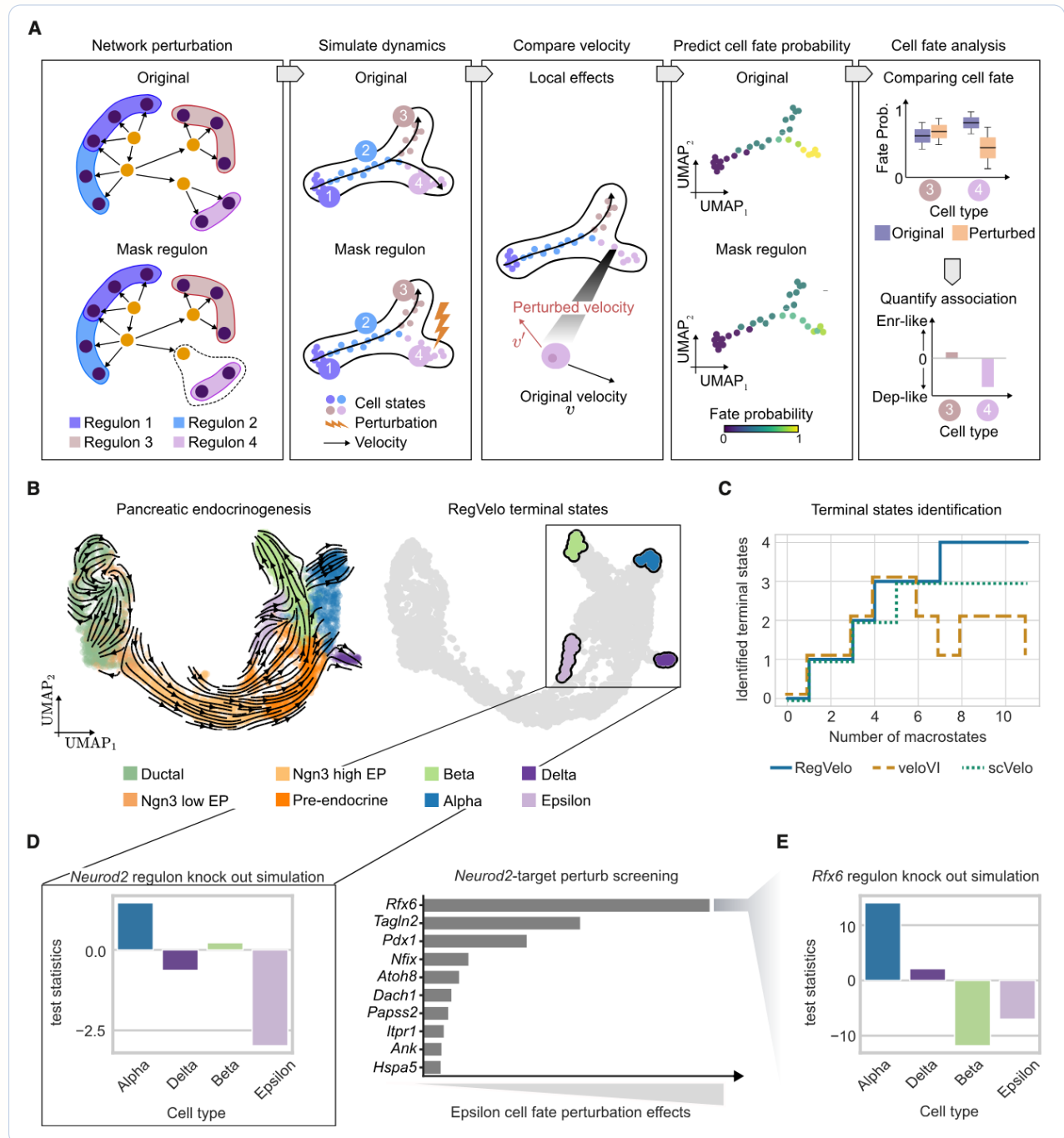


Figure 2: Main text Fig. 2A–E. Perturbation simulation in pancreatic endocrinogenesis. RegVelo uses regulatory perturbations to quantify local velocity shifts and CellRank-derived fate changes, recovering known and candidate endocrine lineage regulators.

Interpretation

Figure note 5.1

This figure is the clearest example of how RegVelo turns a learned GRN into an actionable perturbation model. The important output is not just a list of correlated genes, but the predicted change in terminal fate probability after perturbing a regulon.

5.3 Human hematopoiesis and regulatory motifs

The hematopoiesis experiment tests RegVelo in a system where classical velocity assumptions are often violated. The model uses a prior GRN informed by single-cell multi-omics and ChIP-seq resources. After inte-

grating RegVelo velocities with CellRank, the analysis recovers five terminal states and known hematopoietic transitions.

A key biological result is recovery of the well-known *GATA1-SPI1* toggle-switch motif. Perturbing the *GATA1* regulon is predicted to deplete erythroid fate and enrich monocyte-like fate, while perturbing *SPI1* produces the opposite pattern. This demonstrates how RegVelo can recover regulatory antagonism between lineage programs.

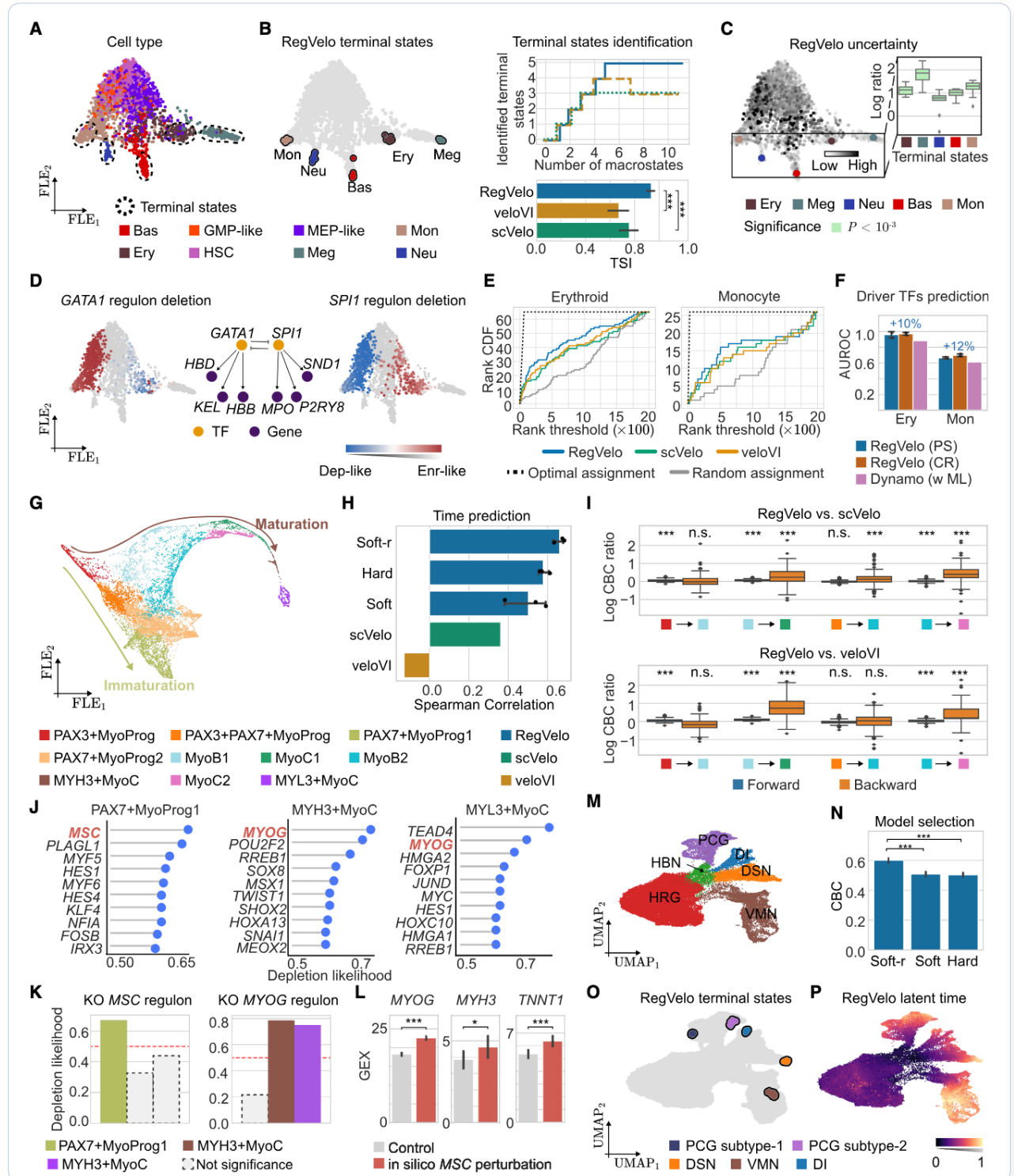


Figure 3: Main text Fig. 3A–P. Regulatory dynamics across hematopoiesis, embryonic myogenesis, and hindbrain development. The most important panels for the hematopoiesis story are the terminal-state recovery, uncertainty map, *GATA1/SPI1* regulon perturbations, and driver prediction results.

How to read Fig. 3

Figure note 5.2

Focus first on panels A–F for hematopoiesis. They show that RegVelo can infer terminal states, uncertainty, and lineage-specific regulator effects in a system where standard velocity methods struggle. The later panels broaden the claim to additional developmental settings.

5.4 Zebrafish neural crest development

The most biologically rich part of the paper is the zebrafish neural crest analysis. Neural crest cells generate multiple lineages, including pigment cells, mesenchymal cells, Schwann cells, sensory neurons, and enteric neurons. RegVelo is trained using Smart-seq3 and shared gene expression/chromatin accessibility information to infer regulatory dynamics during this developmental process.

The model recovers terminal fates and identifies lineage drivers. A major advantage is that RegVelo can prioritize early-expressed drivers that may be missed by simple correlation with terminal fate. This matters because early transcription factors can set up a lineage program long before terminal marker genes become high.

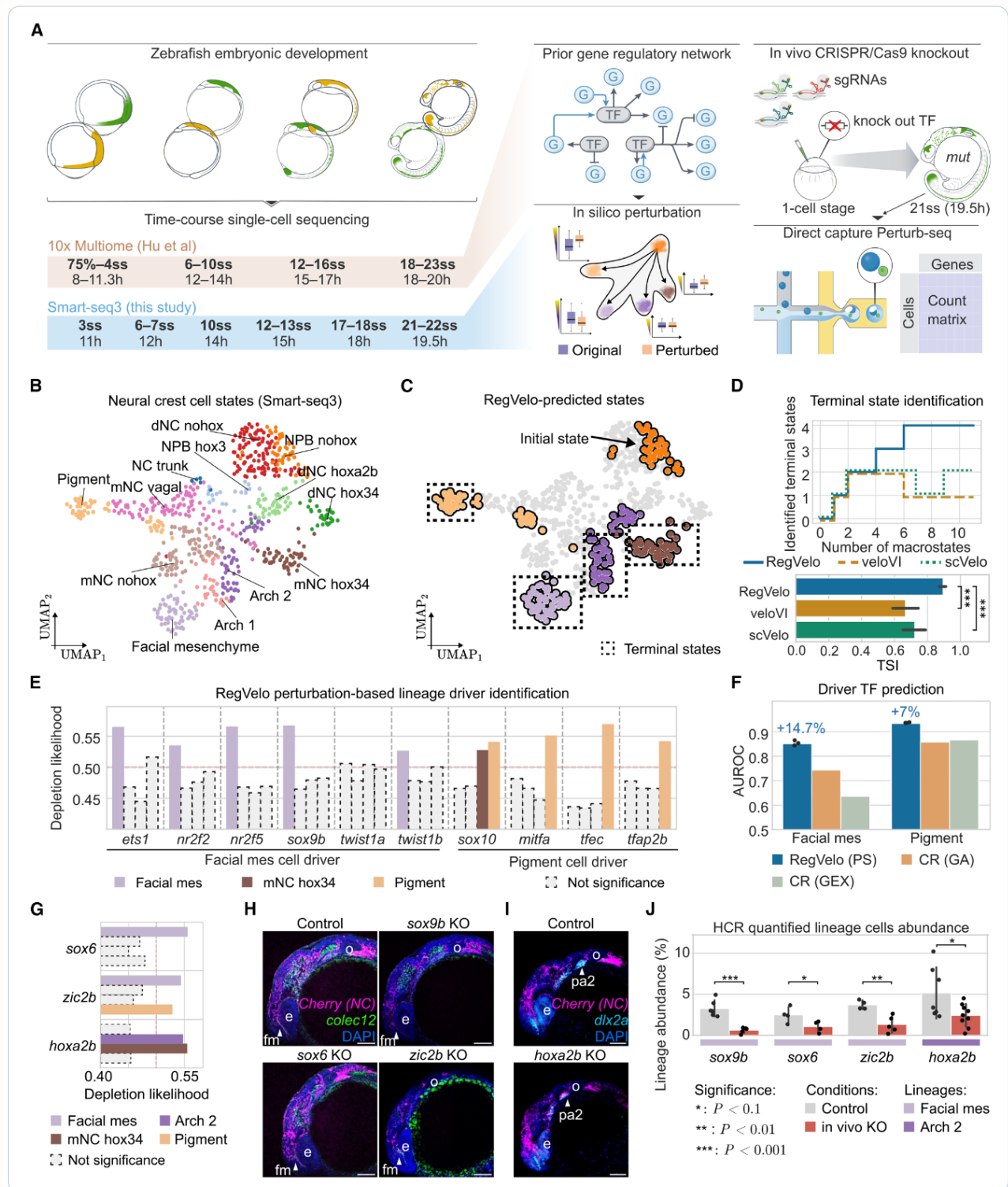


Figure 4: Main text Fig. 4A–N. RegVel analysis of zebrafish neural crest development. Use this figure to show how the model predicts early and lineage-specific regulators from a regulatory dynamical system rather than only from terminal-state expression correlation.

Why this result matters

Figure note 5.3

This is where RegVel's regulatory formulation becomes most important. Early lineage drivers may not be highly expressed in terminal cells, so ranking genes only by expression-fate correlation can miss them. RegVel can detect them because the prior GRN and learned edge weights connect early regulators to downstream fate programs.

5.5 In vivo Perturb-seq validation

The paper validates RegVelo's in silico predictions with CRISPR-Cas9 knockout and direct-capture Perturb-seq in zebrafish. The authors design guide RNAs for candidate transcription factors, generate F0 crispants, and profile neural crest-derived cells. They then compare experimental lineage shifts with RegVelo's simulated perturbation effects.

The experimental results recapitulate known relationships, such as *mitfa* knockout depleting pigment lineage and *nr2f2* knockout depleting mesenchymal fate. RegVelo's density-change likelihood is designed to match these perturbation-induced shifts in terminal cell-state density.

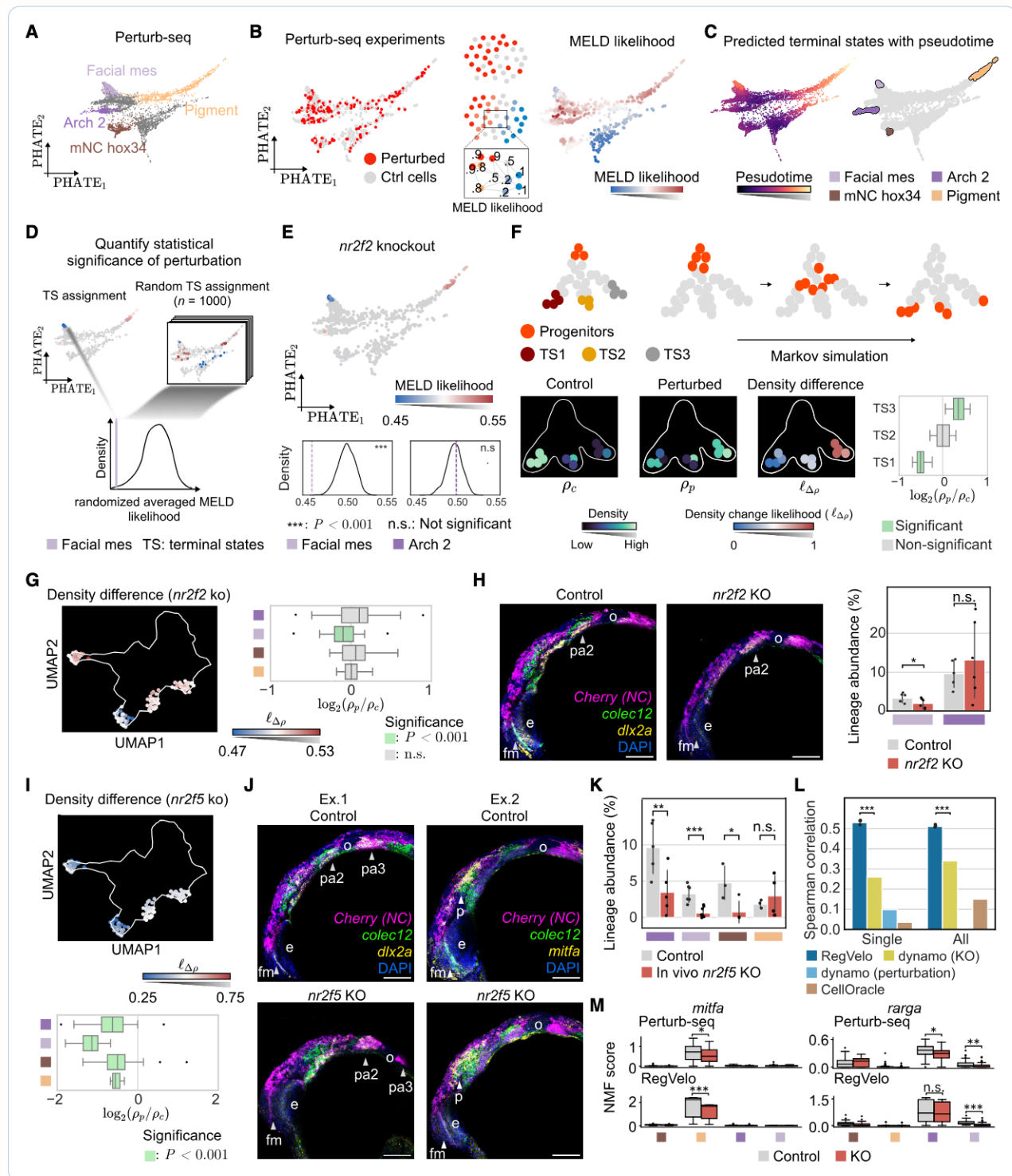


Figure 5: Main text Fig. 5A–M. In vivo Perturb-seq validation of RegVelo perturbation predictions. This figure connects simulated regulon perturbations to experimentally observed lineage shifts in zebrafish neural crest development.

Interpretation

Figure note 5.4

This is an important validation because it moves beyond computational plausibility. RegVelo is evaluated against real perturbation outcomes, so the paper can argue that its simulated regulatory perturbations are biologically actionable rather than merely descriptive.

5.6 *tfec* as an early pigment lineage driver

RegVelo predicts *bhlhe40*, *tfec*, and *mitfa* among the top pigment regulons. The known pigment marker program appears downstream of *tfec*, including targets such as *cdkn1ca*, *atp6ap2*, *slc45a2*, *lrmda*, and *psen2*. The biological interpretation is that *tfec* helps establish pigment fate early in neural crest development.

The paper validates this prediction using HCR imaging and Perturb-seq. *tfec* knockout reduces *sox10/slca45a2* co-expressing pigment-lineage cells, and combinatorial perturbation with related pigment regulators can produce albino-like phenotypes.

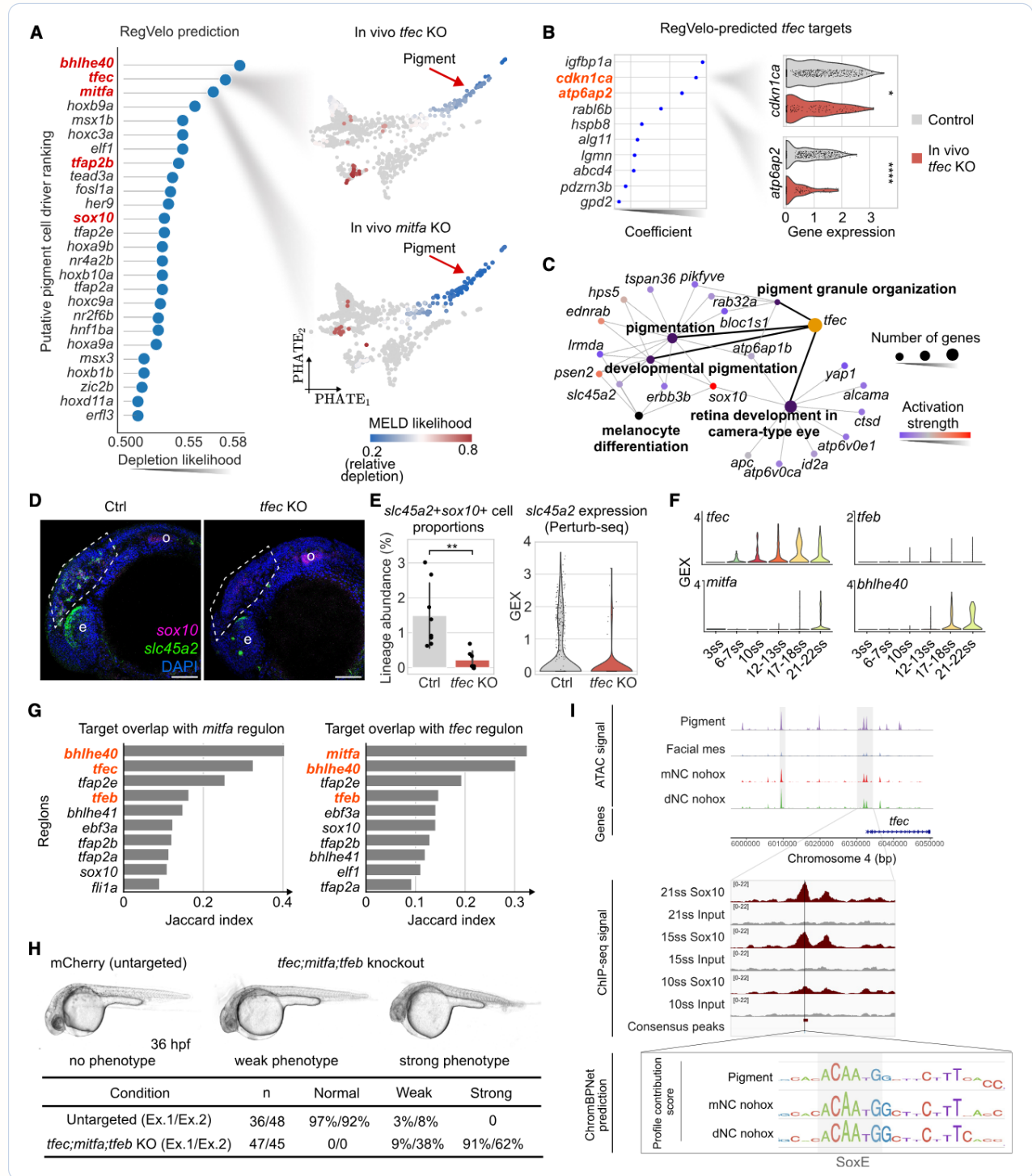


Figure 6: Main text Fig. 6A–H. RegVelo identifies *tfec* as an early pigment lineage driver. The figure combines computational regulon prediction with imaging and Perturb-seq validation.

Why *tfec* is interesting

Figure note 5.5

tfec is not simply a late pigment marker. RegVelo predicts it as an early regulator that helps initiate the pigment program. This illustrates the model's ability to identify drivers that act before terminal fate markers are fully established.

5.7 *elf1* as a pro-pigment regulator

The final biological story is *elf1*. RegVelo predicts *elf1* as a top regulator of pigment fate. Density simulation predicts that *elf1* perturbation reduces pigment terminal-state density and shifts cells toward mesenchymal fates. The in vivo Perturb-seq data support this prediction: *elf1* knockout significantly depletes pigment lineage and downregulates pigment-related genes such as *mitfa* and *sox10*.

The paper further validates this with HCR imaging and explores the regulatory context of *elf1*. The result is especially interesting because many ETS-family factors are associated with mesenchymal programs, whereas *elf1* is presented here as a pro-pigment, anti-mesenchymal regulator.

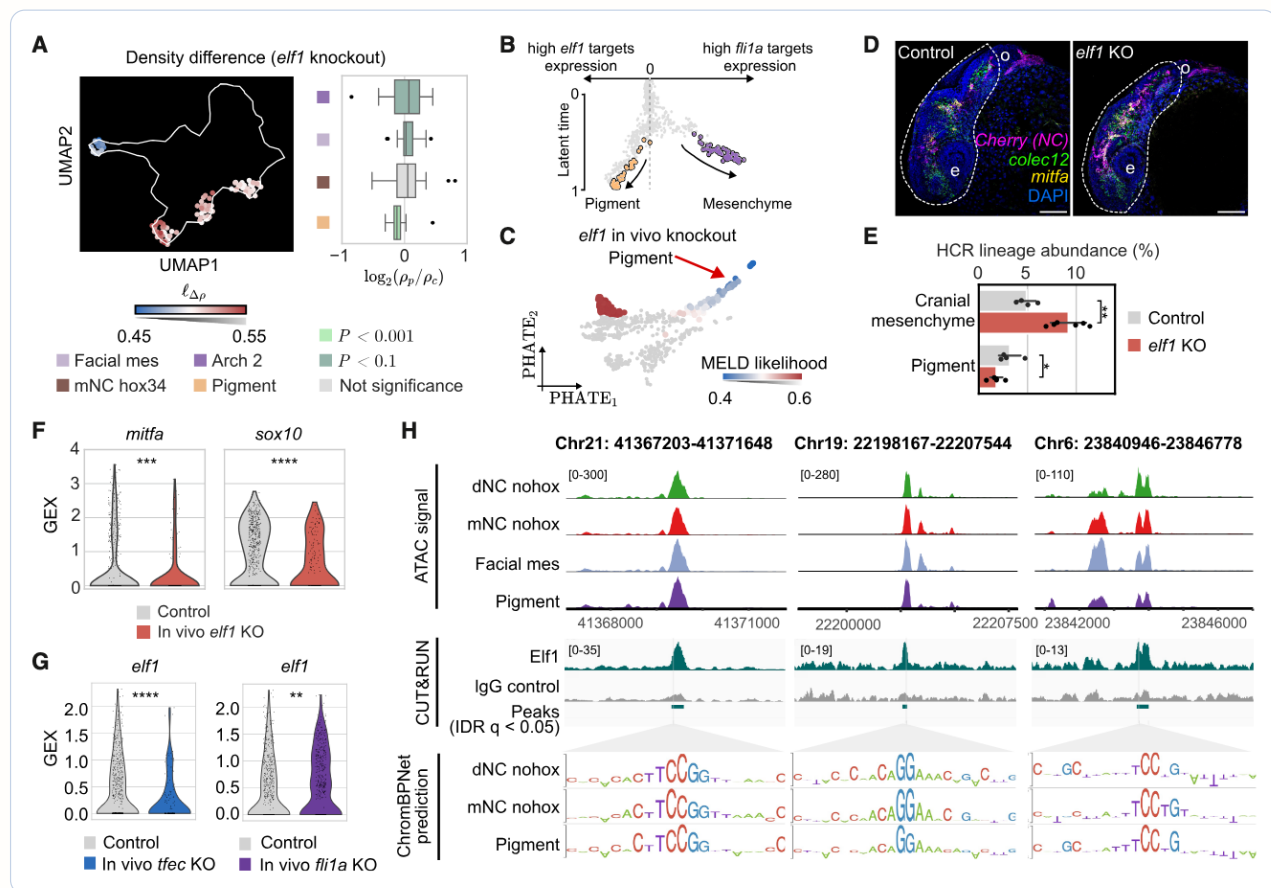


Figure 7: Main text Fig. 7A–G. RegVelo discovers *elf1* as a pro-pigment regulator. This figure is the strongest example of a new biological hypothesis generated by RegVelo and validated experimentally.

Why this is the strongest biological result

Figure note 5.6

The *elf1* story combines model prediction, in silico perturbation, Perturb-seq validation, HCR imaging, and regulatory interpretation. It is the clearest demonstration that RegVelo can generate testable and experimentally supported hypotheses about fate regulation.

6 Core mathematical intuition

6.1 Why couple genes through a GRN?

Classical RNA velocity assumes that each gene follows its own kinetic process. This is mathematically convenient, but it misses the fact that transcription of one gene is often controlled by transcription factors encoded by other genes. RegVelo introduces coupling through the regulatory matrix W .

From independent genes to regulatory systems

Intuition 6.1

In standard velocity, each gene asks: is my spliced RNA increasing or decreasing? In RegVelo, each gene also asks: which regulators are controlling my transcription rate, and how does this regulatory input change across cell states?

6.2 Why use a Bayesian generative model?

Velocity inference is uncertain because single-cell data are sparse and noisy. A Bayesian generative formulation provides posterior distributions rather than only point estimates. This allows RegVelo to quantify uncertainty in velocities and regulatory edges.

6.3 Why integrate with CellRank?

Velocity is local. Fate is long-term. RegVelo estimates the local vector field, while CellRank turns this field into a Markov chain and computes long-term fate probabilities. The two components answer different but complementary questions:

- **RegVelo:** what local regulatory dynamics generate movement?
- **CellRank:** where will cells probably end up if these dynamics are followed?

6.4 Why perturb regulons instead of gene expression directly?

Perturbing a gene's expression manually can create an artificial local shift. Perturbing a regulon changes the underlying regulatory circuit, then lets the dynamical system evolve under that altered circuit. This better represents how transcription factor knockouts can have nonlinear downstream effects over developmental time.

7 Strengths of the paper

- **Strong conceptual integration:** RegVelo unifies RNA velocity, GRN inference, uncertainty estimation, and perturbation simulation.
- **Mechanistic improvement over velocity:** transcription is modeled as regulated and time-dependent rather than gene-independent and constant.
- **Actionable perturbation analysis:** the model can simulate regulon depletion and predict fate probability shifts.
- **Integration with CellRank:** local velocity inference is connected to terminal-state fate mapping.
- **Rich biological validation:** predictions are tested with CRISPR-Cas9 knockout, Perturb-seq, HCR imaging, ChIP-seq, and CUT&RUN-style regulatory evidence.
- **New biological insight:** the paper identifies *tfec* as an early pigment driver and *elf1* as a pro-pigment regulator in zebrafish neural crest development.

- **Uncertainty awareness:** posterior velocity and GRN uncertainty help distinguish reliable from uncertain dynamic and regulatory predictions.

8 Limitations and critical reading

Critical reading

RegVelo is powerful, but its output depends on the quality of the prior GRN, the fit of the dynamical model, the validity of the perturbation simulation, and the extent to which in silico regulon masking reflects real biological perturbation.

8.1 Dependence on prior GRN quality

RegVelo can refine prior regulatory networks, but the prior still matters. If key regulatory edges are absent from the prior, or if many false edges are included, the model may learn an incomplete or biased regulatory system. The paper partly addresses this through sparsity constraints and prior refinement, but prior construction remains a major practical issue.

8.2 Perturbation simulation is not identical to real knockout

In silico perturbation masks a regulon in the learned model. Real CRISPR knockout can produce compensatory effects, incomplete editing, maternal transcript effects, off-target effects, and changes in cell viability. Therefore, RegVelo perturbations are best treated as hypothesis generation unless validated experimentally.

8.3 GRN inference is difficult to validate

A predicted edge may reflect direct regulation, indirect correlation, or shared upstream control. ChIP-seq and CUT&RUN help validate binding, but binding does not always imply functional regulation. Thus, regulatory interpretation still requires caution.

8.4 ODE assumptions may not fit all systems

RegVelo models transcript dynamics through coupled ODEs with neural parameterization. This is flexible, but still an approximation. Systems with strong non-transcriptional regulation, spatial constraints, cell-cell signaling, or abrupt fate decisions may require additional modalities.

9 What I can reuse for my own research

Implications for my work

For pathology-omics and perturbation-aware disease modeling, the most reusable idea is to connect a learned regulatory or causal network to a dynamical system, then use probabilistic fate mapping to predict how perturbations shift endpoint phenotypes.

Potential reusable ideas:

- **Regulatory velocity:** learn a direction field whose dynamics are constrained by a prior regulatory graph.
- **Network perturbation instead of feature perturbation:** simulate interventions by changing regulatory edges or regulons, not only by altering expression values.
- **Local-to-global perturbation effect:** quantify both immediate vector-field shifts and long-term fate probability changes.

- **Uncertainty-aware driver discovery:** rank drivers using both effect size and uncertainty of inferred regulatory edges.
- **Pathology-omics extension:** connect histology-derived cell states, spatial omics, and gene regulatory networks to model transitions toward tumor progression, immune escape, or treatment resistance.

10 Possible extensions

10.1 Spatial RegVelo

Spatial transcriptomics could provide neighborhood and signaling priors. A spatial version of RegVelo could model how local tissue context modifies transcriptional dynamics and fate decisions.

10.2 Perturb-seq-guided RegVelo

Perturb-seq data could be used not only for validation but also during training. This would allow the model to learn regulatory dynamics that are constrained by observed perturbation responses.

10.3 Multi-omics regulatory dynamics

Chromatin accessibility, transcription factor motif activity, protein abundance, and metabolic labeling could be integrated to improve regulatory and kinetic inference. This would be especially useful in systems where RNA velocity alone is weak or misleading.

10.4 Cancer-state transition modeling

In cancer, RegVelo-like modeling could be used to infer regulatory circuits driving transitions between proliferative, invasive, immune-evasive, drug-tolerant, and metastatic states. Perturbation simulation could then nominate candidate therapeutic regulators.

11 Questions to ask while reading

Questions for further thought

1. How sensitive are RegVelo predictions to the prior GRN construction method?
2. Does the model distinguish direct regulation from indirect regulatory correlation?
3. When simulated perturbation disagrees with Perturb-seq, is the failure due to model dynamics, GRN inference, or experimental noise?
4. How robust are fate probability shifts to CellRank parameters and terminal-state definitions?
5. Can RegVelo scale to very large atlas-level datasets with many regulators and targets?
6. How should one validate early drivers that are no longer expressed in terminal cells?

12 My final evaluation

Table 1: Personal evaluation of the paper.

Criterion	Evaluation
Novelty	Very high. RegVelo connects RNA velocity, GRN inference, Bayesian uncertainty, CellRank fate mapping, and regulon perturbation simulation in one framework.
Biological value	Very high. The zebrafish neural crest analysis generates experimentally validated hypotheses, especially for <i>tfec</i> and <i>elf1</i> in pigment fate regulation.
Technical clarity	Strong but complex. The core idea is clear, but the full model combines neural ODEs, variational inference, GRN priors, uncertainty estimation, and downstream fate mapping.
Validation	Strong. The paper uses simulations, cell-cycle benchmarks, pancreatic endocrinogenesis, hematopoiesis, zebrafish neural crest, CRISPR knockouts, Perturb-seq, HCR imaging, and regulatory binding assays.
Main caveat	The method depends on prior GRN quality and on how faithfully in silico regulon masking approximates biological perturbation.
Reading rating	★ 4.8/5

Key takeaway

RegVelo is best understood as a regulatory extension of RNA velocity: it does not only infer where cells are moving, but also models which regulatory circuits may drive that movement and what happens when those circuits are perturbed.